



9674 - Turbulent Dissipation Regions Revealed

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	Taurus_Filament	MIRI Medium Resolution Spectroscopy	(1) Taurus_Filament
	11	Taurus_Filament-bkg	MIRI Medium Resolution Spectroscopy	(11) Taurus_Filament-bkg
	2	Rho_Oph	MIRI Medium Resolution Spectroscopy	(2) Rho_Oph
	12	Rho_Oph-bkg	MIRI Medium Resolution Spectroscopy	(12) Rho_Oph-bkg
	3	Musca_Filament	MIRI Medium Resolution Spectroscopy	(3) Musca_filament
	13	Musca_Filament-bkg	MIRI Medium Resolution Spectroscopy	(13) Musca_filament-bkg
	4	B68	MIRI Medium Resolution Spectroscopy	(4) B68
	14	B68-bkg	MIRI Medium Resolution Spectroscopy	(14) B68-bkg

ABSTRACT

Turbulence is fundamental to ISM dynamics and hence to star formation in molecular clouds. Turbulent energy flows in a cascade from large scales, driven by galactic rotation, massive young stars, supernovae, and cloud-cloud collisions, down to small scales where it dissipates. But small scales where viscosity becomes significant and dissipation occurs are poorly understood. Estimates of the dissipation scale range from 10 to 100s of au, requiring the angular resolution provided by JWST even in the nearest star-forming regions. Recent JWST Cy2 observations (proj. JWST-GO-03050.001-A) of H₂ yielded the first images of turbulent dissipation regions, 100-300 au in size, in the Taurus molecular cloud boundary. The emission from H₂ in the J = 3 level demands temperatures $T > 500$ K and densities of a few hundred /cm³ in regions where the bulk of the H₂ is at $T \sim 10$ K. Such temperatures and densities are also required to produce the otherwise inexplicable widespread presence of CH⁺ in the ISM. The implied small filling factor suggests that turbulent dissipation regions heat the gas intermittently. We propose observing 4 regions in the 17 micron H₂ line with JWST/MIRI to sample different environments within 150 pc of the Sun. The proximity of these regions together with JWST's exquisite angular resolution will enable precise determination of turbulent dissipation mechanisms. Demonstrating that turbulent dissipation is widespread and determining the properties of the dissipative structures are vital for modeling the structure of interstellar molecular clouds, and will illuminate the long standing question of how turbulence influences star formation and thus galaxy evolution.

OBSERVING DESCRIPTION

This proposal requests observation of four molecular cloud boundaries to image regions of turbulent dissipation occurring there. The four clouds have different turbulent to thermal linewidth ratios and are in different environments, but are all within 150 pc of the sun, thus allowing the angular resolution of JWST to resolve the ~few hundred au size of such condensations such as observed in the linear edge boundary of Taurus by Goldsmith et al. (2005). By imaging and resolving regions of turbulent dissipation we will vastly improve the overall understanding of turbulence in interstellar molecular clouds and quantify the energy input and flow from large-scale to the size of the dissipation regions where vorticity results in dissipation, heating, and emission of the H₂ S(1) 17.04 micron line.

We propose using MIRI/MRS Channel 3 sub-band C. We will employ a 4x4 mosaicing of the MRS sub-band C pointings to cover the S(1) line in a combined FoV of 24"x25". A FOV of this size is essential to obtain reasonable sampling of the dissipation regions as can be seen in images in Goldsmith et al. (2005). In each of the MRS pointings, we adopt a detector setup of 12 groups per integration and 2-pt dither pattern reaching an exposure time of 573 sec. We have identified appropriate point sources for MRS target acquisition.

Proposal 9674 - Targets - Turbulent Dissipation Regions Revealed

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	Taurus_Filament	RA: 04 50 34.0000 (72.6416667d) Dec: +25 18 30.00 (25.30833d) Equinox: J2000 Comments: Category=ISM Description=[Dark interstellar clouds, Dense interstellar clouds, Diffuse interstellar clouds, Interstellar clouds, Interstellar molecules] Extended=YES		
(2)	Rho_Oph	RA: 16 26 47.5000 (246.6979167d) Dec: -24 48 30.00 (-24.80833d) Equinox: J2000 Comments: Category=ISM Description=[Dark interstellar clouds, Dense interstellar clouds, Diffuse interstellar clouds, Interstellar clouds, Interstellar molecules] Extended=YES		
(3)	Musca_filament	RA: 12 27 20.0000 (186.8333333d) Dec: -72 20 0.00 (-72.33333d) Equinox: J2000 Comments: Category=ISM Description=[Dark interstellar clouds, Dense interstellar clouds, Diffuse interstellar clouds, Interstellar clouds, Interstellar molecules] Extended=YES		
(4)	B68	RA: 17 22 38.2000 (260.6591667d) Dec: -23 51 24.00 (-23.85667d) Equinox: J2000 Comments: Category=ISM Description=[Dark interstellar clouds, Dense interstellar clouds, Diffuse interstellar clouds, Interstellar clouds, Interstellar molecules] Extended=YES		
(11)	Taurus_Filament-bkg	RA: 04 48 28.4000 (72.1183333d) Dec: +25 35 12.20 (25.58672d) Equinox: J2000 Comments: Category=ISM Description=[Dark interstellar clouds, Dense interstellar clouds, Diffuse interstellar clouds, Interstellar clouds, Interstellar molecules] Extended=YES		
(12)	Rho_Oph-bkg	RA: 16 26 47.5000 (246.6979167d) Dec: -24 30 0.00 (-24.50000d) Equinox: J2000 Comments: Category=ISM Description=[Dark interstellar clouds, Dense interstellar clouds, Diffuse interstellar clouds, Interstellar clouds, Interstellar molecules] Extended=YES		
(13)	Musca_filament-bkg	RA: 12 25 30.0000 (186.3750000d) Dec: -71 41 30.00 (-71.69167d) Equinox: J2000 Comments: Category=ISM Description=[Dark interstellar clouds, Dense interstellar clouds, Diffuse interstellar clouds, Interstellar clouds, Interstellar molecules] Extended=YES		

Proposal 9674 - Targets - Turbulent Dissipation Regions Revealed

(14)	B68-bkg	RA: 17 22 38.2000 (260.6591667d) Dec: -23 49 34.00 (-23.82611d) Equinox: J2000
Comments: Category=ISM Description=[Dark interstellar clouds, Dense interstellar clouds, Diffuse interstellar clouds, Interstellar clouds, Interstellar molecules]		

Proposal 9674 - Observation 1 - Turbulent Dissipation Regions Revealed

Observation	Proposal 9674, Observation 1: Taurus_Filament												Fri Mar 13 17:04:06 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
Diagnostics	(Visit 1:1) Warning (Form): Data Excess over middle threshold												
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(1)	Taurus_Filament	RA: 04 50 34.0000 (72.6416667d) Dec: +25 18 30.00 (25.30833d) Equinox: J2000										
	Comments: Category=ISM Description=[Dark interstellar clouds, Dense interstellar clouds, Diffuse interstellar clouds, Interstellar clouds, Interstellar molecules] Extended=YES												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel			Simultaneous Imaging			Imager Subarray			Grating Wheel Direction		
		Channel 3			YES			FULL			Allow Auto Reorder		
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order		
	4	4	-20.0		-20.0		0.0		0.0		DEFAULT		
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				EXTENDED SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F770W	SLOWR1	16	1	1	Dither 1	4	4	1528.955	
	1	LONG(C)	MRSLONG		FASTR1	140	1	1	Dither 1	4	4	1554.022	
	1	LONG(C)	MRSSHORT		FASTR1	140	1	1	Dither 1	4	4	1554.022	

Proposal 9674 - Observation 11 - Turbulent Dissipation Regions Revealed

Observation	Proposal 9674, Observation 11: Taurus_Filament-bkg												Fri Mar 13 17:04:06 GMT 2026
	Diagnostic Status: Warning												
Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(Visit 11:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous			
	(11)	Taurus_Filament-bkg	RA: 04 48 28.4000 (72.1183333d) Dec: +25 35 12.20 (25.58672d) Equinox: J2000										
	Comments: Category=ISM Description=[Dark interstellar clouds, Dense interstellar clouds, Diffuse interstellar clouds, Interstellar clouds, Interstellar molecules] Extended=YES												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel			Simultaneous Imaging			Imager Subarray		Grating Wheel Direction			
		Channel 3			YES			FULL		Allow Auto Reorder			
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				EXTENDED SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F770W	SLOWR1	16	1	1	Dither 1	4	4	1528.955	
	1	LONG(C)	MRSLONG		FASTR1	140	1	1	Dither 1	4	4	1554.022	
	1	LONG(C)	MRSSHORT		FASTR1	140	1	1	Dither 1	4	4	1554.022	

Proposal 9674 - Observation 2 - Turbulent Dissipation Regions Revealed

Observation	Proposal 9674, Observation 2: Rho_Oph												Fri Mar 13 17:04:06 GMT 2026
	Diagnostic Status: Warning												
Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(Visit 2:1) Warning (Form): Data Excess over middle threshold												
	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous		
	(2)	Rho_Oph	RA: 16 26 47.5000 (246.6979167d) Dec: -24 48 30.00 (-24.80833d) Equinox: J2000										
	Comments: Category=ISM Description=[Dark interstellar clouds, Dense interstellar clouds, Diffuse interstellar clouds, Interstellar clouds, Interstellar molecules] Extended=YES												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction		
		Channel 3				YES			FULL		Allow Auto Reorder		
Mosaic	Rows	Columns	Row Overlap %			Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order	
	4	4	-20.0			-20.0		0.0		0.0		DEFAULT	
Dithers	#	Dither Type					Optimized For			Direction			
	1	4-Point					EXTENDED SOURCE			NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F770W	SLOWR1	16	1	1	Dither 1	4	4	1528.955	
	1	LONG(C)	MRSLONG		FASTR1	140	1	1	Dither 1	4	4	1554.022	
	1	LONG(C)	MRSSHORT		FASTR1	140	1	1	Dither 1	4	4	1554.022	

Proposal 9674 - Observation 12 - Turbulent Dissipation Regions Revealed

Observation	Proposal 9674, Observation 12: Rho_Oph-bkg												Fri Mar 13 17:04:06 GMT 2026
	Diagnostic Status: Warning												
Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(Visit 12:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous		
	(12)	Rho_Oph-bkg	RA: 16 26 47.5000 (246.6979167d) Dec: -24 30 0.00 (-24.50000d) Equinox: J2000										
	Comments: Category=ISM Description=[Dark interstellar clouds, Dense interstellar clouds, Diffuse interstellar clouds, Interstellar clouds, Interstellar molecules] Extended=YES												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction		
		Channel 3				YES			FULL		Allow Auto Reorder		
Dithers	#	Dither Type					Optimized For			Direction			
	1	4-Point					EXTENDED SOURCE			NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F770W	SLOWR1	16	1	1	Dither 1	4	4	1528.955	
	1	LONG(C)	MRSLONG		FASTR1	140	1	1	Dither 1	4	4	1554.022	
	1	LONG(C)	MRSSHORT		FASTR1	140	1	1	Dither 1	4	4	1554.022	

Proposal 9674 - Observation 3 - Turbulent Dissipation Regions Revealed

Observation	Proposal 9674, Observation 3: Musca_Filament												Fri Mar 13 17:04:06 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
Diagnostics	(Visit 3:1) Warning (Form): Data Excess over middle threshold												
	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous			
	(3)	Musca_filament	RA: 12 27 20.0000 (186.8333333d) Dec: -72 20 0.00 (-72.33333d) Equinox: J2000										
	Comments: Category=ISM Description=[Dark interstellar clouds, Dense interstellar clouds, Diffuse interstellar clouds, Interstellar clouds, Interstellar molecules] Extended=YES												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel			Simultaneous Imaging			Imager Subarray			Grating Wheel Direction		
		Channel 3			YES			FULL			Allow Auto Reorder		
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order		
	4	4	-20.0		-20.0		0.0		0.0		DEFAULT		
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				EXTENDED SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F770W	SLOWR1	16	1	1	Dither 1	4	4	1528.955	
	1	LONG(C)	MRSLONG		FASTR1	140	1	1	Dither 1	4	4	1554.022	
	1	LONG(C)	MRSSHORT		FASTR1	140	1	1	Dither 1	4	4	1554.022	

Proposal 9674 - Observation 13 - Turbulent Dissipation Regions Revealed

Observation	Proposal 9674, Observation 13: Musca_Filament-bkg												Fri Mar 13 17:04:06 GMT 2026
	Diagnostic Status: Warning												
Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(Visit 13:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous		
	(13)	Musca_filament-bkg	RA: 12 25 30.0000 (186.3750000d) Dec: -71 41 30.00 (-71.69167d) Equinox: J2000										
Comments: Category=ISM Description=[Dark interstellar clouds, Dense interstellar clouds, Diffuse interstellar clouds, Interstellar clouds, Interstellar molecules] Extended=YES													
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction		
		Channel 3				YES			FULL		Allow Auto Reorder		
Dithers	#	Dither Type					Optimized For			Direction			
	1	4-Point					EXTENDED SOURCE			NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F770W	SLOWR1	16	1	1	Dither 1	4	4	1528.955	
	1	LONG(C)	MRSLONG		FASTR1	140	1	1	Dither 1	4	4	1554.022	
	1	LONG(C)	MRSSHORT		FASTR1	140	1	1	Dither 1	4	4	1554.022	

Proposal 9674 - Observation 4 - Turbulent Dissipation Regions Revealed

Observation	Proposal 9674, Observation 4: B68												Fri Mar 13 17:04:06 GMT 2026
	Diagnostic Status: Warning												
Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(Visit 4:1) Warning (Form): Data Excess over middle threshold												
	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous		
	(4)	B68	RA: 17 22 38.2000 (260.6591667d) Dec: -23 51 24.00 (-23.85667d) Equinox: J2000										
	Comments:												
	Category=ISM Description=[Dark interstellar clouds, Dense interstellar clouds, Diffuse interstellar clouds, Interstellar clouds, Interstellar molecules]												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel			Simultaneous Imaging			Imager Subarray			Grating Wheel Direction		
		Channel 3			YES			FULL			Allow Auto Reorder		
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order		
	4	4	-20.0		-20.0		0.0		0.0		DEFAULT		
Dithers	#	Dither Type					Optimized For			Direction			
	1	4-Point					EXTENDED SOURCE			NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F770W	SLOWR1	16	1	1	Dither 1	4	4	1528.955	
	1	LONG(C)	MRSLONG		FASTR1	140	1	1	Dither 1	4	4	1554.022	
	1	LONG(C)	MRSSHORT		FASTR1	140	1	1	Dither 1	4	4	1554.022	

Proposal 9674 - Observation 14 - Turbulent Dissipation Regions Revealed

Observation	Proposal 9674, Observation 14: B68-bkg												Fri Mar 13 17:04:06 GMT 2026
	Diagnostic Status: Warning												
Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(Visit 14:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous		
	(14)	B68-bkg	RA: 17 22 38.2000 (260.6591667d) Dec: -23 49 34.00 (-23.82611d) Equinox: J2000										
	Comments: Category=ISM Description=[Dark interstellar clouds, Dense interstellar clouds, Diffuse interstellar clouds, Interstellar clouds, Interstellar molecules]												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction		
		Channel 3				YES			FULL		Allow Auto Reorder		
Dithers	#	Dither Type					Optimized For			Direction			
	1	4-Point					EXTENDED SOURCE			NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F770W	SLOWR1	16	1	1	Dither 1	4	4	1528.955	
	1	LONG(C)	MRSLONG		FASTR1	140	1	1	Dither 1	4	4	1554.022	
	1	LONG(C)	MRSSHORT		FASTR1	140	1	1	Dither 1	4	4	1554.022	